NAME: GOKULAKRISHNAN AS **ROLL NO :250701203**

EXPERIMENT 16

Exp No:16 Experiment to understand EDA-Quantitative and Qualitative analysis.

AIM:

To understand EDA-Quantitative and Qualitative analysis.

DESCRIPTION:

Understand the importance of EDA-Quantitative and Qualitative analysis.

CODE:

```
import seaborn as sns

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

%matplotlib inline

# Read uploaded dataset

tips = pd.read_csv('/content/tip.csv')

print(tips)

print("\n")

print(tips.head())

print("\n")

# Distribution plot with KDE

sns.displot(tips.total_bill, kde=True)

plt.show()
```

```
# Distribution plot without KDE
sns.displot(tips.total_bill, kde=False)
plt.show()
# Joint plot
sns.jointplot(x=tips.tip, y=tips.total_bill)
plt.show()
# Joint plot with regression
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="reg")
plt.show()
# Joint plot with hex
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="hex")
plt.show()
# Pair plot
sns.pairplot(tips)
plt.show()
# Time value counts
print(tips.time.value_counts())
print("\n")
# Pair plot with hue = time
sns.pairplot(tips, hue='time')
plt.show()
# Pair plot with hue = day
sns.pairplot(tips, hue='day')
plt.show()
```

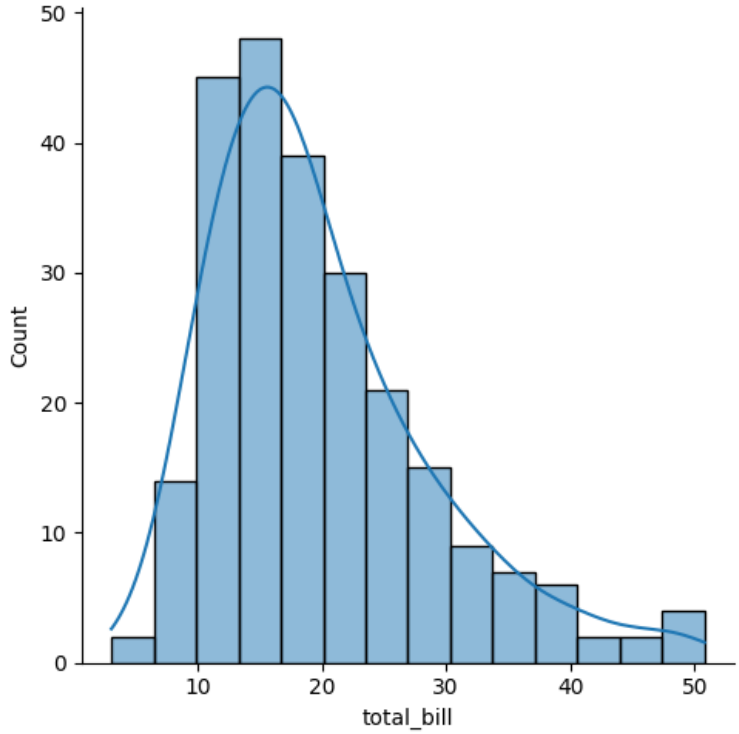
```
# Heatmap
sns.heatmap(tips.corr(numeric_only=True), annot=True)
plt.show()
# Boxplot total_bill
sns.boxplot(x=tips.total_bill)
plt.show()
# Boxplot tip
sns.boxplot(x=tips.tip)
plt.show()
# Countplot day
sns.countplot(x=tips.day)
plt.show()
# Countplot sex
sns.countplot(x=tips.sex)
plt.show()
# Pie chart of sex
tips.sex.value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.show()
# Bar chart of sex
tips.sex.value_counts().plot(kind='bar')
plt.show()
# Dinner day countplot
sns.countplot(x=tips[tips.time == 'Dinner']['day'])
plt.show()
```

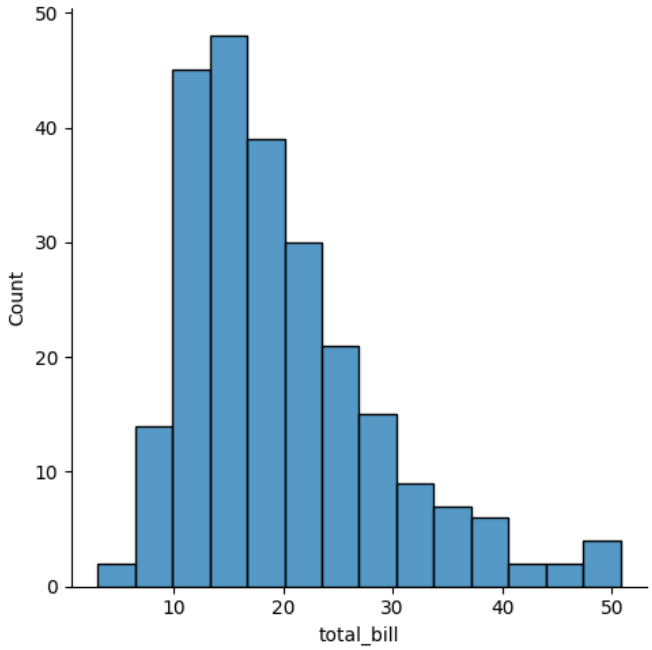
OUTPUT:

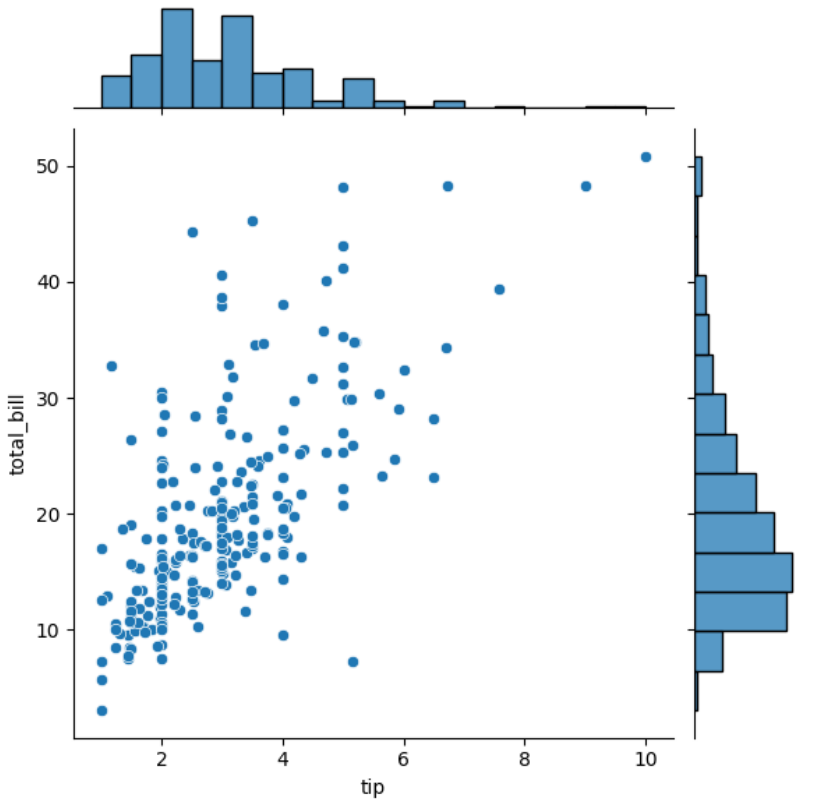
```
     total_bill   tip     sex smoker   day    time  size
0         16.99  1.01  Female     No   Sun  Dinner     2
1         10.34  1.66    Male     No   Sun  Dinner     3
2         21.01  3.50    Male     No   Sun  Dinner     3
3         23.68  3.31    Male     No   Sun  Dinner     2
4         24.59  3.61  Female     No   Sun  Dinner     4
..          ...   ...     ...    ...   ...     ...   ...
239       29.03  5.92    Male     No   Sat  Dinner     3
240       27.18  2.00  Female    Yes   Sat  Dinner     2
241       22.67  2.00    Male    Yes   Sat  Dinner     2
242       17.82  1.75    Male     No   Sat  Dinner     2
243       18.78  3.00  Female     No  Thur  Dinner     2

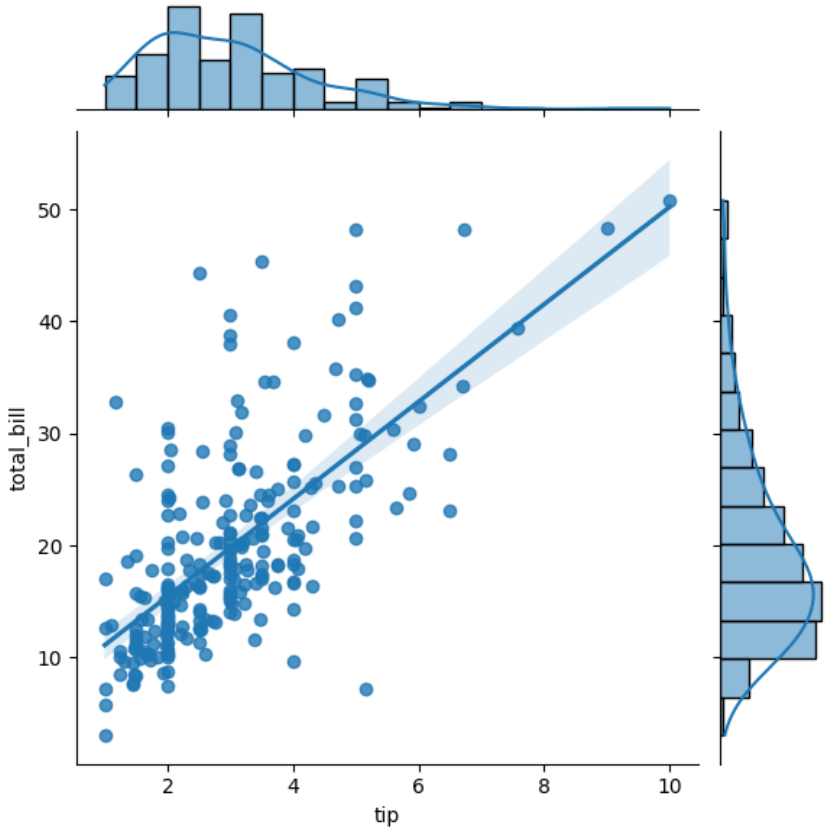
[244 rows x 7 columns]
```

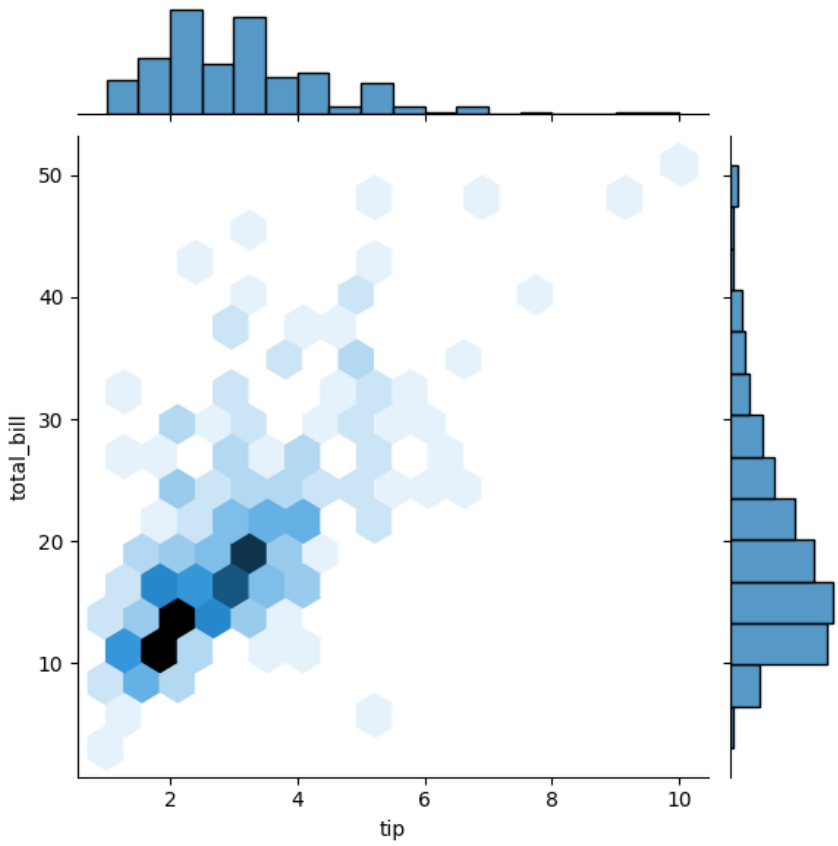
```
     total_bill   tip     sex smoker  day    time  size
  0       16.99  1.01  Female     No  Sun  Dinner     2
  1       10.34  1.66    Male     No  Sun  Dinner     3
  2       21.01  3.50    Male     No  Sun  Dinner     3
  3       23.68  3.31    Male     No  Sun  Dinner     2
  4       24.59  3.61  Female     No  Sun  Dinner     4
```

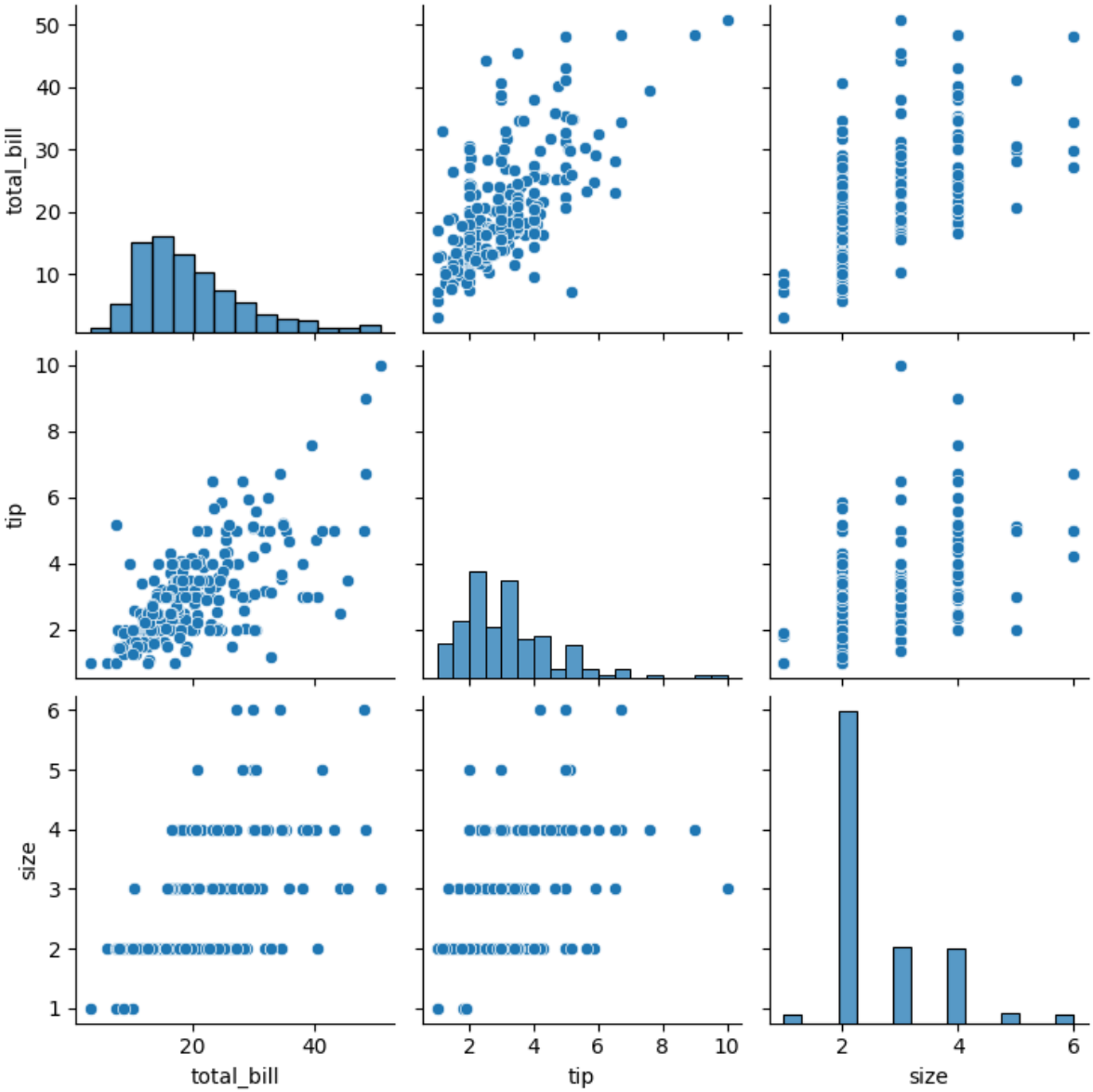

Count
total_bill
0
10
20
30
40
50


Count
total_bill
0
10
20
30
40
50

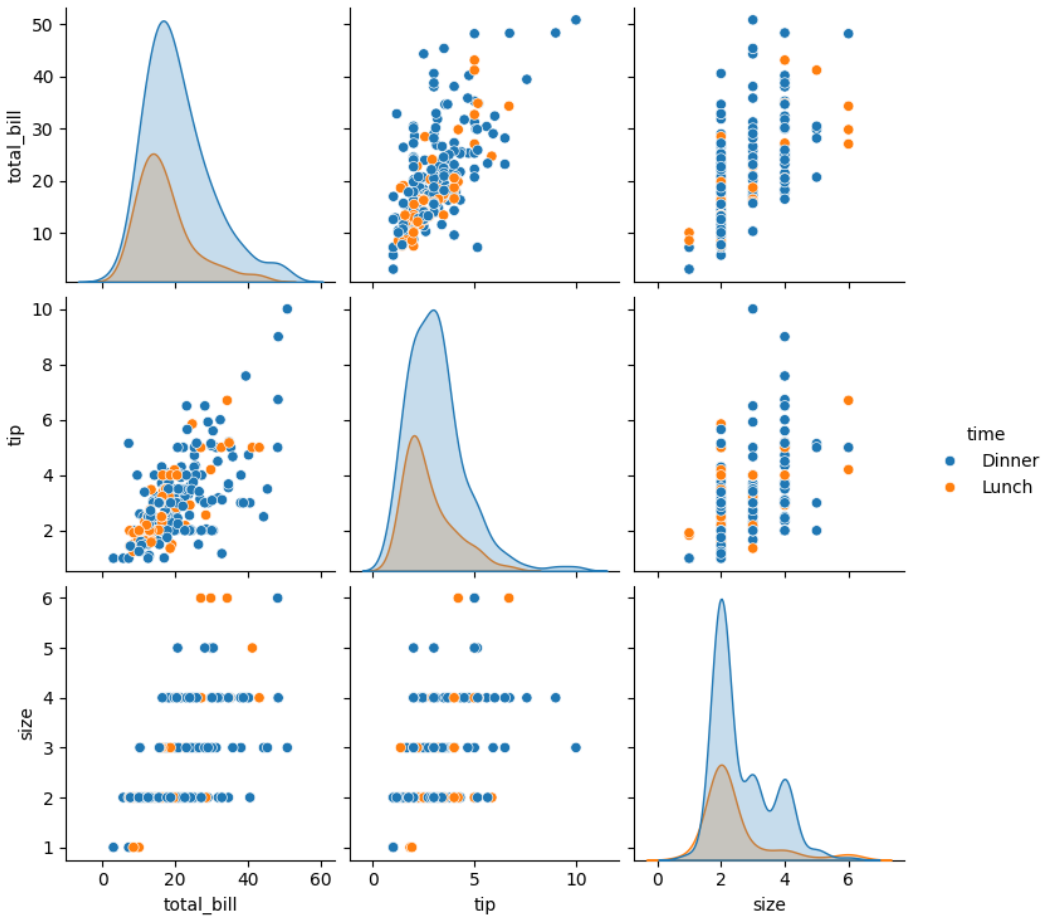

total_bill
tip
50
40
30
20
10
2
4
6
8
10

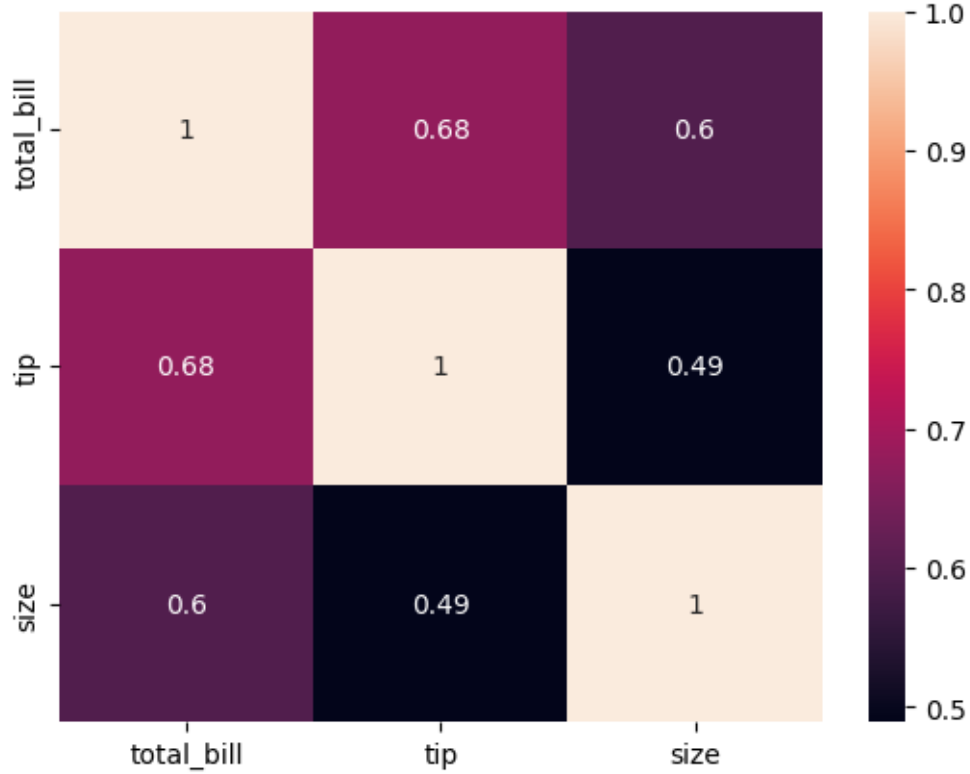

total_bill
tip
50
40
30
20
10
2
4
6
8
10

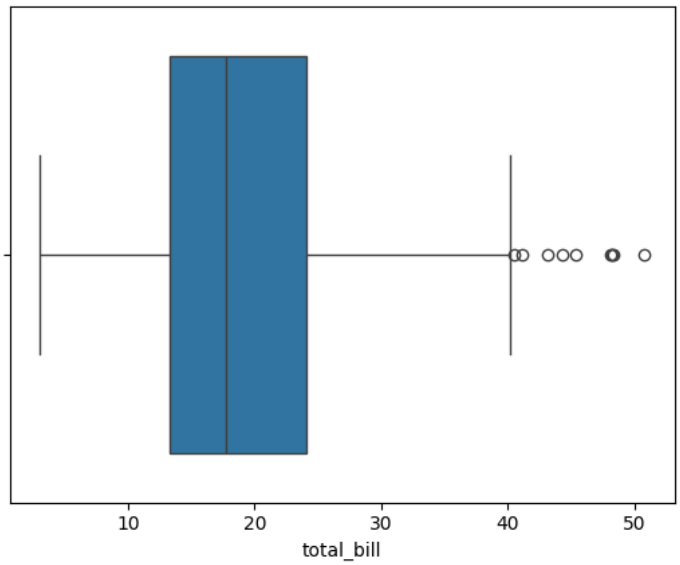

total_bill
tip
50
40
30
20
10
2
4
6
8
10

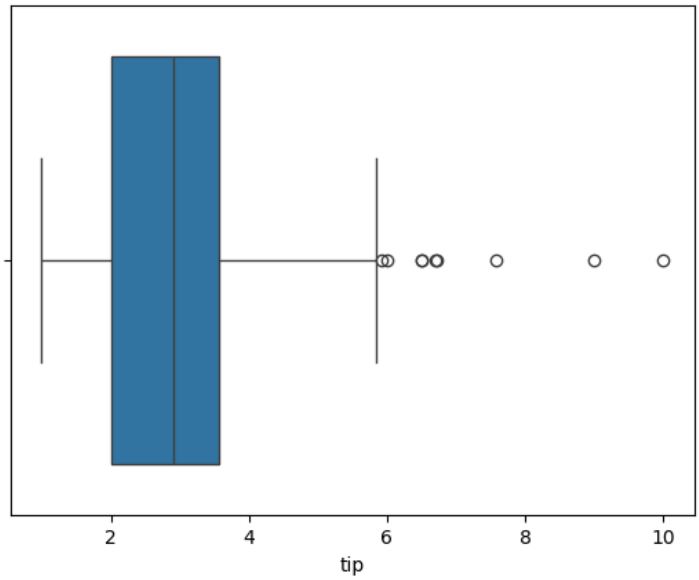


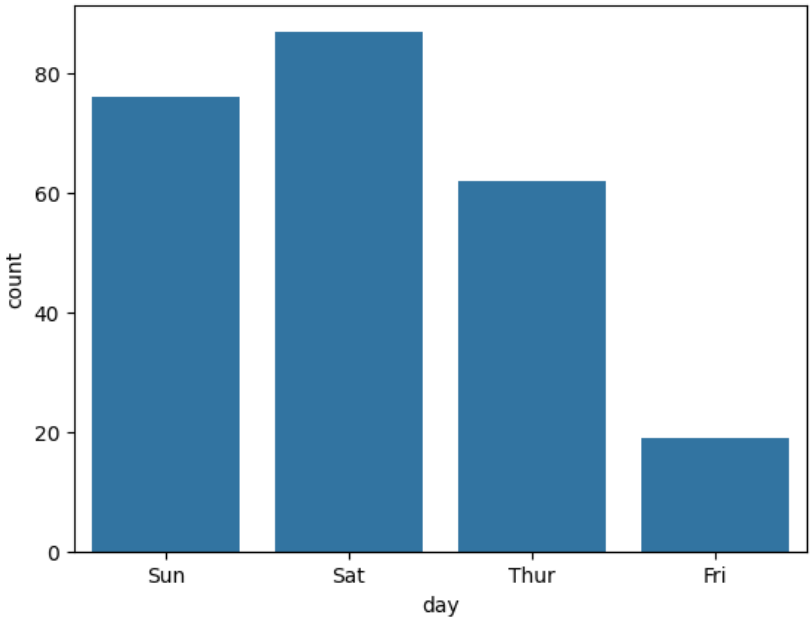

```
time
Dinner    176
Lunch      68
Name: count, dtype: int64
```

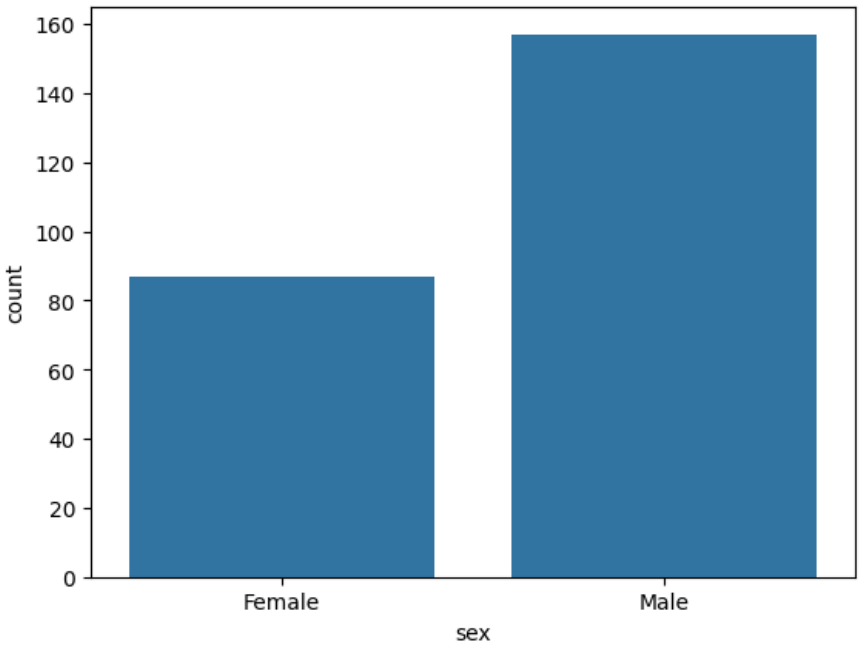

total_bill
tip
size
time
Dinner
Lunch

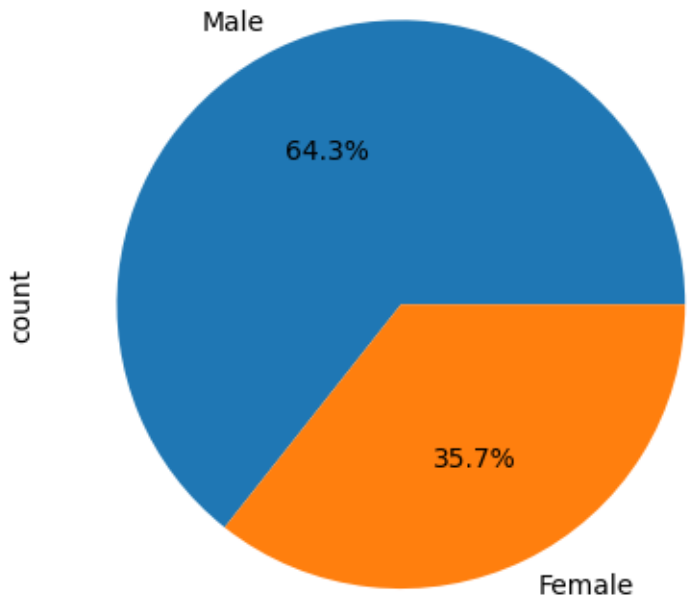

total_bill
tip
size
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
1.0
0.9
0.8
0.7
0.6
0.5

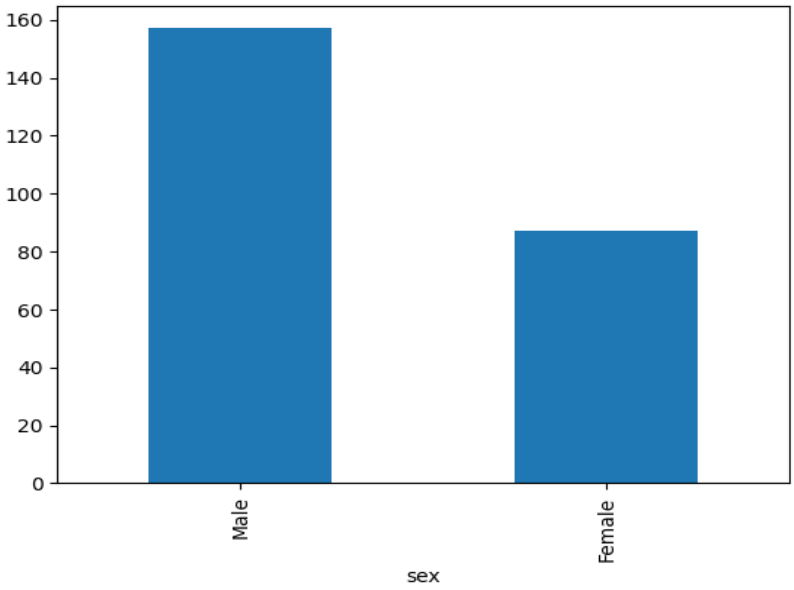

10
20
30
40
50
total_bill

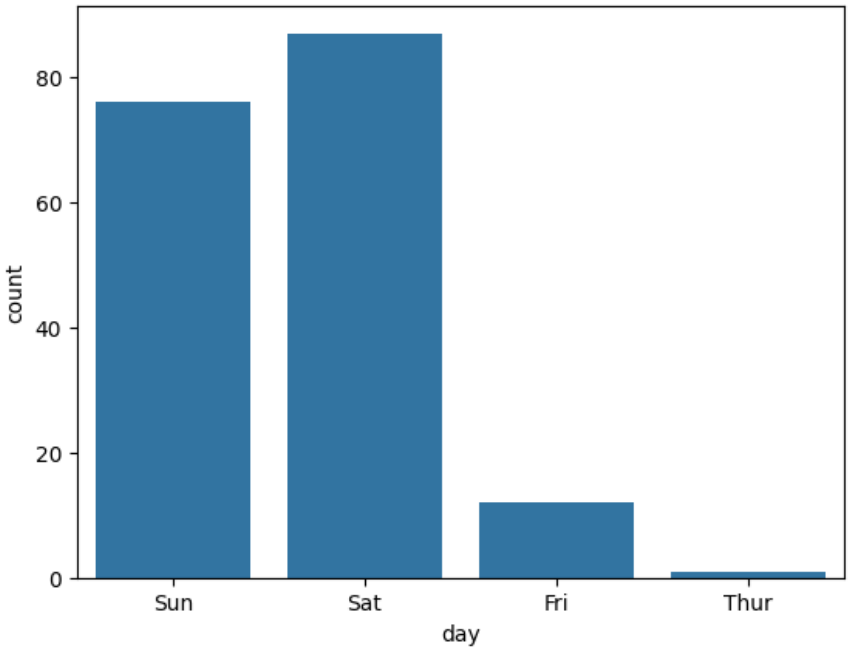

2
4
6
8
10
tip


0
20
40
60
80
count
Sun
Sat
Thur
Fri
day


160
140
120
100
80
60
40
20
0
count
Female
Male
sex


Male
64.3%
count
35.7%
Female





RESULT:

Thus, Exploratory Data Analysis (EDA) was performed successfully using Pandas, Seaborn, and Matplotlib. Various plots were generated to analyze quantitative and qualitative data in the tips dataset.